

ARTICLE TYPE

STRUCTURAL FOUNDATIONS OF SUPPLY CHAIN RESILIENCE: NATIONAL PRODUCTIVE COMPLEXITY AS A MACRO-LEVEL DETERMINANT UNDER TARIFF AND GEOPOLITICAL SHOCKS

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Abstract

Supply chain resilience research has concentrated predominantly on firm-level mechanisms — agility, redundancy, and relational competencies — while largely neglecting the structural national contexts in which firms are embedded. This paper challenges this firm-centric orthodoxy by arguing that a country's productive complexity, as captured by the Economic Complexity Index (ECI), constitutes a fundamental macro-level determinant of supply chain resilience that is analytically prior to, and partially constitutive of, the firm-level capabilities emphasized in extant theory. Drawing on a panel dataset of 130 countries over 2012–2022, we exploit the staggered tariff escalation rounds of the 2018–2019 US–China trade war as a natural experiment to estimate the causal effect of national productive complexity on three distinct dimensions of supply chain resilience: global value chain (GVC) linkage stability, export portfolio recovery speed, and trade partner diversification under stress. Using two-way fixed effects difference-in-differences estimation with propensity score matching as a robustness check, we find consistent and robust evidence that countries with higher ECI experience significantly smaller disruptions to their GVC embeddedness ($\beta = 0.092, p < 0.01$), recover their export volumes faster following tariff shocks ($\beta = 0.161, p < 0.01$), and diversify to alternative trading partners more successfully ($\beta = 0.114, p < 0.01$). These effects are amplified by the product-space density captured in the Complexity Outlook Index and attenuated under extreme geopolitical risk. Our findings advance supply chain management theory by establishing a country-level structural construct that complements firm-level resilience mechanisms, with important implications for supply chain network design, sourcing strategy, and public policy.

KEYWORDS

supply chain resilience, economic complexity, global value chains, tariff shocks, geopolitical risk, difference-in-differences, product space, country-level analysis

1 | INTRODUCTION

The global supply chain landscape has undergone a seismic structural transformation in the years following the 2018–2019 US–China trade war. The imposition of sweeping tariff schedules across thousands of product categories, subsequently followed by the COVID-19 pandemic, the Russia–Ukraine conflict, and renewed geopolitical fracturing of established trade relationships, has exposed fundamental vulnerabilities in supply chains optimized for efficiency under conditions of sustained

globalization. For supply chain managers, the challenge is no longer merely one of managing dyadic supplier relationships or building organizational agility: it is one of identifying which national environments offer structural conditions that permit rapid adaptation, rerouting, and recovery when external shocks sever established trade flows. Yet the academic field of supply chain management has provided limited guidance on precisely this question.

The dominant paradigm in supply chain resilience research is unambiguously firm-centric. Since Christopher and Peck's (2004) foundational articulation of supply chain resilience, the field has converged

Abbreviations: ANA, anti-nuclear antibodies; APC, antigen-presenting cells; IRF, interferon regulatory factor.

on a set of firm-level and dyadic constructs: agility, robustness, redundancy, flexibility, and relational competencies (Wieland & Wallenburg, 2013; Brandon-Jones et al., 2014; Ambulkar et al., 2015; Durach et al., 2015). While this literature has produced important insights, it systematically understates the role of national structural context. A firm's capacity to absorb, adapt to, and recover from a supply chain disruption is not solely a function of its internal capabilities or its relationships with immediate trading partners — it is also profoundly shaped by the broader productive ecosystem of the country in which it is embedded. A Vietnamese electronics manufacturer and a Bangladeshi garment producer may have identical internal supply chain management practices, yet face radically different resilience prospects when a geopolitical shock disrupts their export markets, because the productive ecosystems they inhabit — the depth of their national knowhow, the diversity of their export capabilities, the connectivity of their product space — are fundamentally different.

This paper directly challenges the firm-centric orthodoxy by proposing that national productive complexity — as operationalized through the Economic Complexity Index (ECI) developed at Harvard's Growth Lab (Hidalgo & Hausmann, 2009; Hausmann et al., 2014) — constitutes a structural macro-level determinant of supply chain resilience that is both analytically distinct from, and empirically prior to, the firm-level mechanisms emphasized in current theory. The ECI captures the diversity and sophistication of productive capabilities embedded in a country's export portfolio, reflecting the accumulated stock of distributed productive knowledge — what Hidalgo and Hausmann (2009) call 'knowhow' — that enables an economy to sustain and reconfigure its participation in global production networks. We argue that countries with high ECI possess a structural resilience advantage: their firms have access to deeper pools of complementary capabilities, more diverse and substitutable export pathways, and a more connected product space that allows rapid pivoting to adjacent products and markets when disruptions strike established trade flows. This advantage is not reducible to firm-level management practices; it is an emergent property of the national productive ecosystem itself.

To test this argument, we exploit a powerful natural experiment. The United States Trade Representative (USTR) implemented four staggered rounds of tariff escalation against Chinese imports between July 2018 and September 2019, covering a cumulative value of approximately \$550 billion in goods at rates ranging from 7.5% to 25%. These tariff rounds were largely exogenous to the structural characteristics of third-party countries — they were imposed based on US policy objectives vis-à-vis China, not on the economic characteristics of Vietnam, Germany, Mexico, or Bangladesh. This exogeneity allows us to use a difference-in-differences (DiD) framework in which countries more heavily exposed to the US-China trade nexus (through either competitive or supply chain linkages) serve as the treated group, while less-exposed countries serve as controls. Crucially, we interact treatment exposure with ECI to test whether higher national productive complexity buffers countries against the disruptions induced by the tariff shock.

Our empirical analysis draws on a panel dataset covering 130 countries from 2012 to 2022, combining data from the Harvard Growth Lab's Atlas of Economic Complexity, the OECD Trade in Value Added (TiVA) database, the United Nations Comtrade database, the UNCTAD TRAINS bilateral tariff database, the World Bank's World Development Indicators, the World Governance Indicators, and the Caldara and Iacoviello (2022) Geopolitical Risk Index. We examine three distinct dimensions of supply chain resilience as dependent variables: (1) the stability of a country's forward GVC linkage share following the tariff shock, (2) the speed with which a country's export portfolio recovers to pre-shock volumes, and (3) the degree to which a country successfully diversifies its trade partners under shock conditions. Across all three dimensions and multiple robustness checks — including propensity score matching, placebo tests using pre-period pseudo-shocks, exclusion of resource-rich and OECD countries, and stacked DiD event-study specifications — we find consistent and statistically significant evidence that higher ECI substantially and causally buffers supply chain resilience outcomes.

The contributions of this paper are threefold. First, theoretically, we challenge the firm-centric view of supply chain resilience by establishing a country-level structural construct — national productive complexity — that is both analytically distinct from existing firm-level constructs and empirically powerful in explaining resilience heterogeneity across countries. This contribution opens a new theoretical avenue for supply chain management research by connecting the discipline to the emerging literature on economic complexity, productive knowhow, and structural economic transformation. Second, methodologically, we introduce a natural experiment design to the JSCM literature on resilience, providing causal identification of the ECI-resilience relationship that correlational studies cannot achieve. Third, practically, our findings offer supply chain managers a new macro-level screening tool for sourcing country selection under tariff and geopolitical risk, and provide policymakers with evidence for investments in productive diversification as a resilience-building strategy.

The remainder of this paper is organized as follows. Section 2 reviews the extant supply chain resilience literature, critiques its firm-centric foundations, and develops our theoretical framework and hypotheses. Section 3 describes our data, variable operationalization, and empirical methodology. Section 4 presents the main results and moderation analyses. Section 5 discusses the theoretical and practical implications. Section 6 concludes.

2 | THEORETICAL BACKGROUND AND HYPOTHESES

2.1 | Supply Chain Resilience: The Dominant Firm-Centric View

Supply chain resilience (SCR) has been defined in multiple, partially overlapping ways since the concept gained academic prominence following

the disruptions of the early 2000s. Christopher and Peck's (2004) seminal definition — the ability of a supply chain to return to its original state or move to a new, more desirable state after being disturbed — established resilience as fundamentally an adaptive property of a system that has experienced an exogenous shock. Subsequent definitional refinements have emphasized proactive preparation (Tang, 2006), dynamic capabilities (Teece et al., 1997; Ambulkar et al., 2015), and the temporal dynamics of absorption, adaptation, and recovery (Ponomarov & Holcomb, 2009; Wieland & Wallenburg, 2013).

What is striking across this definitional landscape is the consistent focus on the firm or the dyadic relationship as the unit of analysis. The most influential theoretical frameworks in SCR research have concentrated on: (1) structural characteristics of supply chains — complexity, density, criticality (Craighead et al., 2007; Choi & Krause, 2006); (2) firm capabilities — sensing, learning, reconfiguration (Ambulkar et al., 2015; Durach et al., 2015); (3) relational competencies — information sharing, collaborative flexibility, trust (Wieland & Wallenburg, 2013; Brandon-Jones et al., 2014); and (4) strategic approaches — redundancy, diversification, postponement (Tang, 2006; Sheffi, 2005). These are all firm-level or dyadic constructs, operationalized and measured at the level of the individual firm or firm-to-firm relationship.

The institutional and national context in which these firm-level mechanisms operate receives, at best, marginal theoretical attention. Blackhurst et al. (2005) note that supply chain vulnerabilities have geographic dimensions, and Ivanov and Dolgui (2020) acknowledge that supply chain viability is shaped by macro-level environmental conditions. Yet neither of these observations has been translated into a theoretical framework that systematically incorporates national structural characteristics as determinants of SCR. The implicit assumption in most SCR research is that national context is either a control variable — something to be held constant while examining firm-level predictors — or an exogenous environmental shock, rather than a structural enabler or disabler of resilient supply chain behavior.

This firm-centric view has produced important insights, but it systematically misspecifies the determinants of resilience heterogeneity across countries. A multinational supply chain strategist designing sourcing networks under the new tariff and geopolitical risk environment cannot be guided solely by firm-level resilience research: they need a macro-level framework that explains why some countries consistently produce more resilient supply chain nodes than others, irrespective of firm-level management practice.

2.2 | The Structural National Context as a Neglected Determinant

The neglect of national structural context in SCR research stands in sharp contrast to several adjacent literatures that have long recognized the decisive importance of where firms are located. Porter's (1990) Diamond Model established that national competitive advantage — and thus the productive capabilities available to firms — is fundamentally shaped by factor conditions, related and supporting industries, and the

depth of domestic rivalry. Dunning's (1998) OLI framework made locational advantages central to foreign direct investment theory. North's (1990) institutional economics demonstrated that country-level institutional frameworks shape the transaction costs and capability-building environments available to firms. And the global value chain literature (Gereffi et al., 2005; Kano et al., 2020) has increasingly recognized that a firm's position in, and resilience within, a global production network depends on the broader national productive ecosystem in which it is embedded.

From these adjacent literatures, a clear theoretical premise emerges: the productive capabilities available to firms in a given country are not purely the result of those firms' internal decisions and resource accumulations. They are partially constituted by the broader productive ecosystem — the density of complementary capabilities, the depth of the domestic knowledge base, the variety of related and supporting industries, and the accumulated productive knowhow that is embedded in the national economy and available as a shared resource to firms operating within it. A firm in Germany can draw on an extraordinarily dense ecosystem of precision engineering capabilities, advanced material science knowhow, and sophisticated logistics infrastructure that firms in less complex economies simply cannot access, regardless of their internal strategic decisions.

This observation points to a fundamental limitation in the SCR literature's firm-centric approach: it misattributes to firm-level management what is partially a structural property of national productive ecosystems. A firm's ability to reconfigure its supply chain following a disruption depends not only on its internal capabilities and relational assets, but on the availability of complementary capabilities in its national environment — alternative suppliers, adjacent product competencies, logistics infrastructure, technical knowhow in related industries — that can be mobilized rapidly in the face of disruption. Where these structural conditions exist, resilience is easier to achieve and less dependent on anticipatory managerial foresight. Where they are absent, even the most strategically sophisticated firm may be structurally constrained in its resilience response.

2.3 | Economic Complexity Theory and Supply Chain Adaptive Capacity

Economic complexity theory, developed by Hidalgo and Hausmann (2009) and formalized in the Atlas of Economic Complexity (Hausmann et al., 2014), provides precisely the theoretical constructs needed to operationalize the structural national characteristics that SCR research has neglected. The core claim of complexity economics is that the wealth and growth trajectory of national economies is best explained not by their factor endowments or their policy environments per se, but by the diversity and sophistication of the productive capabilities — or 'knowhow' — embedded in their economies and expressed through their export portfolios.

The ECI is computed from bilateral trade data using an iterative method that simultaneously estimates country complexity from product

complexity and product complexity from country complexity, converging on an eigenvector solution that reflects the full information content of the country-product bipartite trade network (Hidalgo & Hausmann, 2009). A high ECI score indicates that a country exports many different products (diversity), and that those products tend to be exported by few other countries (ubiquity of producers is low), implying that producing them requires rare and sophisticated productive capabilities. The product-level analogue, the Product Complexity Index (PCI), captures the sophistication and rarity of the capabilities required to produce a given good.

Two additional constructs from this framework are particularly relevant for our argument. The Complexity Outlook Index (COI) measures the density of products in a country's product space that are adjacent to — i.e., require overlapping capabilities with — the country's current export portfolio but have not yet been captured. A high COI indicates that a country has many opportunities for capability-based diversification into new products without requiring entirely new types of productive knowledge. The Product Space itself (Hidalgo et al., 2007) is a network of products connected by their proximity — the degree to which the capabilities required to produce them overlap — and visualizes the paths through which countries can realistically diversify their productive structures.

The implications of these constructs for supply chain resilience are theoretically powerful. A country with high ECI commands a deep and diverse stock of productive knowhow, a large and varied export portfolio, and a dense product space in which many products are capability-adjacent. When a tariff shock or geopolitical disruption closes off a particular export pathway — say, automotive components exports to the United States — a high-ECI country has available to it a wider array of alternative production pathways (adjacent products), alternative export destinations (flexible productive capabilities that serve multiple markets), and a deeper domestic ecosystem of complementary capabilities that facilitate rapid reconfiguration. Conversely, a low-ECI country that has concentrated its productive capabilities in a narrow range of low-complexity commodities or simple manufactures faces structural constraints on reconfiguration: its firms have fewer alternative products to pivot to, its domestic ecosystem provides fewer complementary capabilities, and its export portfolio's concentration means that the closure of a single trade pathway has outsized disruptive effects.

2.4 | Bridging Economic Complexity and Supply Chain Resilience Theory

To formally connect economic complexity theory with supply chain resilience, we draw on a resource-based mechanism that complements, rather than supplants, existing SCR theory. The Resource-Based View (RBV; Barney, 1991; Peteraf, 1993) argues that sustained competitive advantage derives from resources that are valuable, rare, inimitable, and non-substitutable. Applied at the firm level, the RBV has been used to explain why some firms are more resilient than others (Ambulkar et

al., 2015; Brandon-Jones et al., 2014). We extend this logic to the national level, following Dierickx and Cool's (1989) argument that resource stocks are accumulated through sustained investment over time and cannot be rapidly replicated.

We conceptualize national productive complexity — as captured by the ECI — as a national-level resource endowment that is: (1) valuable, because it enables flexible production and export reconfiguration across multiple dimensions; (2) rare, because ECI is highly heterogeneous across countries and represents the accumulated outcome of decades of economic learning and capability building; (3) inimitable, because productive knowhow is tacit, path-dependent, and deeply embedded in national institutional and social structures that cannot be rapidly replicated; and (4) non-substitutable, because no policy instrument or strategic decision can quickly replicate the accumulated knowhow embedded in a complex economy. This conceptualization allows us to argue that the ECI-resilience relationship is not simply a statistical artifact of economic development — controlling for GDP per capita — but represents a theoretically grounded causal mechanism through which productive structural endowments translate into resilience outcomes.

Crucially, we position national productive complexity as a structural enabler of the firm-level resilience mechanisms identified in extant SCR theory. Complex economies produce firms with more diverse and flexible capabilities not because those firms individually possess superior resources, but because they operate within national ecosystems that provide deep pools of complementary knowhow, sophisticated logistics and infrastructure, and diverse networks of capable suppliers and partners. National complexity is thus the structural substrate in which firm-level resilience mechanisms are embedded and from which they derive part of their efficacy. This theoretical claim suggests that the firm-centric SCR literature has systematically underestimated the structural determinants of resilience by conflating what is nationally structural with what is firm-specific.

2.5 | Hypothesis Development

2.5.1 | National Complexity and GVC Linkage Stability

Global value chain (GVC) linkages — the degree to which a country's production is integrated into international production networks through forward (supplying intermediates to foreign producers) and backward (importing foreign intermediates for domestic production) linkages — represent a core dimension of supply chain embeddedness. Disruptions to GVC linkages are among the most consequential consequences of tariff shocks and geopolitical upheaval (Baldwin, 2016; Kano et al., 2020), as they reflect the degree to which a country maintains its position as a node in the global production network.

We argue that countries with higher ECI will experience smaller disruptions to their forward GVC linkages following tariff shocks. The mechanism operates through three pathways. First, high-ECI countries produce a more diverse set of intermediate goods for global value

chains, meaning that disruption to any single trade pathway has a smaller proportional impact on total GVC embeddedness. Second, high-ECI countries' intermediate exports are harder to substitute, because they embody rare and sophisticated capabilities that alternative sourcing locations cannot easily replicate — making their GVC partners more likely to absorb the tariff cost or restructure their arrangements than to exit the relationship entirely. Third, high-ECI countries are better positioned to rapidly redirect their intermediate exports to alternative buyer countries when traditional pathways are closed, because their productive capabilities are recognized as valuable in multiple end markets. Together, these mechanisms predict that national productive complexity will buffer against GVC linkage disruption.

H1: Countries with higher economic complexity experience significantly smaller disruptions to their forward GVC linkage shares following exogenous tariff shocks, relative to less complex economies.

2.5.2 | National Complexity and Export Portfolio Recovery Speed

Beyond the immediate impact of a tariff shock on GVC linkages, supply chain resilience requires the capacity to recover — to restore export volumes and trade flows to pre-shock levels or to redirect them to equivalent alternative destinations (Ponomarov & Holcomb, 2009; Christopher & Peck, 2004). We argue that national productive complexity substantially accelerates the speed of export portfolio recovery following a tariff-induced disruption.

The theoretical mechanism is grounded in the productive knowledge structure of complex economies. When a tariff shock disrupts an established trade pathway, firms in high-ECI countries can draw on a dense ecosystem of related capabilities to rapidly pivot to adjacent products or markets. The Product Space logic (Hidalgo et al., 2007) implies that high-ECI countries have many nearby products — products requiring overlapping capabilities with their current portfolio — to which they can redirect productive resources relatively quickly. Additionally, complex economies typically feature more sophisticated financial systems, logistics infrastructure, and business service ecosystems that reduce the transaction costs of market-switching, enabling faster reorientation of export flows. By contrast, low-ECI economies, often specialized in a narrow range of commodities or low-complexity manufactures, face structural constraints on pivot pathways and higher reconfiguration costs, resulting in slower recovery.

H2: Countries with higher economic complexity recover their export portfolio volumes more rapidly following exogenous tariff shocks, relative to less complex economies.

2.5.3 | National Complexity and Trade Partner Diversification Under Stress

A third critical dimension of supply chain resilience under tariff and geopolitical shocks is the ability to diversify trade partners — to successfully redirect export flows to alternative destination markets when primary markets are disrupted (Tang, 2006). We argue that national productive complexity is a powerful enabler of this trade partner diversification capacity.

The mechanism operates through the recognition of complex economies' exports as valuable in multiple end markets simultaneously. Products with high PCI — characteristic of high-ECI export portfolios — are globally demanded and can be sold in diverse markets, because they embody capabilities that are scarce worldwide. When a primary export market is disrupted by tariffs or geopolitical tensions, firms in high-ECI countries can leverage the global scarcity value of their sophisticated exports to penetrate alternative markets. By contrast, firms in low-ECI economies, often exporting commodity products with near-zero PCI, face intense competition in alternative markets where numerous other low-complexity economies offer identical or near-identical products, making successful diversion far more difficult. Additionally, high-ECI economies' firms are more likely to have pre-existing relationships with multiple end markets — because their sophisticated exports attract global buyer interest — reducing the search and relationship costs of trade partner switching.

H3: Countries with higher economic complexity demonstrate significantly greater trade partner diversification under exogenous tariff shocks, relative to less complex economies.

2.5.4 | The Moderating Role of Product Space Density

While ECI captures the overall level of productive complexity, the Complexity Outlook Index (COI) captures a distinct structural feature: the density of a country's product space relative to its current export capabilities. A high COI indicates that the country has many products in its product space that are nearby to its current capabilities — products it could feasibly produce with modest additional investment in related knowhow — but has not yet captured. This forward-looking product space density represents an additional structural resilience resource: it determines the richness of the adaptive pathways available to a country when disruption forces reconfiguration.

We argue that COI amplifies the ECI-resilience relationship: for a given level of ECI, countries with higher COI have more adjacent product pathways available for reconfiguration, enabling faster and more complete resilience responses. The interaction between ECI and COI captures the combined structural endowment of current capability sophistication and available capability expansion pathways.

H4: Product space density (COI) positively moderates the relationship between economic complexity and supply chain resilience, such that the ECI-resilience relationship is stronger for countries with higher COI.

2.5.5 | The Moderating Role of Geopolitical Risk

The structural resilience advantage conferred by high ECI is not unconditional. We argue that extreme geopolitical risk — involving state-level coercion, sanctions, military conflict, or severe diplomatic rupture — can undermine the mechanisms through which productive complexity translates into supply chain resilience. When geopolitical risk reaches extreme levels, the capacity of high-ECI countries to leverage their sophisticated export capabilities in alternative markets may be constrained by politically motivated trade restrictions, sanctions regimes that block access to financial systems, or reputational and security concerns that deter buyer relationships. In such conditions, structural productive capability is a necessary but not sufficient condition for resilience; geopolitical alignment and regime compatibility become increasingly decisive.

Consistent with Caldara and Iacoviello's (2022) finding that geopolitical risk has non-linear effects on trade and capital flows, we hypothesize that extreme GPR attenuates the ECI-resilience relationship, as the channels through which productive complexity confers resilience — global market access, buyer recognition of capability value, trade partner switching — are progressively closed off by geopolitical barriers.

H5: Geopolitical risk negatively moderates the relationship between economic complexity and supply chain resilience, such that the ECI-resilience relationship is weaker under high geopolitical risk conditions.

3 | RESEARCH DESIGN AND METHODOLOGY

3.1 | The Natural Experiment: US-China Tariff Escalation (2018–2019)

Our empirical strategy exploits the staggered tariff escalation imposed by the United States on Chinese imports between July 2018 and September 2019 as a natural experiment. The USTR implemented four distinct tariff lists, each covering a separate set of HS-8 product categories at rates of 7.5% to 25%: List 1 (\$34 billion, effective July 6, 2018), List 2 (\$16 billion, effective August 23, 2018), List 3 (\$200 billion at 10%, effective September 24, 2018, escalated to 25% in May 2019), and List 4A (\$112 billion at 7.5%, effective September 1, 2019). These tariff rounds created sharp, exogenous, product-category-specific trade disruptions that were driven by US policy objectives vis-à-vis China rather than by the productive characteristics of third-country economies.

The exogeneity of the shock to third-country characteristics is the critical identifying assumption of our design. The product categories selected for tariff imposition reflected US strategic concerns about Chinese industrial policy, technology transfer, and trade practices — as documented in the USTR Section 301 investigation — rather than any assessment of how the productive complexity of Vietnam, Germany, or Mexico would respond. We validate this assumption through the parallel trends test (presented in Appendix D) and through placebo tests using pre-period pseudo-shocks (Table 6), both of which support the identifying assumption.

The mechanism through which the US-China tariff war creates a resilience test for third-country supply chains operates through two primary pathways. First, countries that compete with China in US markets face a trade diversion opportunity: the tariffs make Chinese goods more expensive, potentially creating export opportunities for alternative suppliers. Whether countries can capture these opportunities — rapidly redirecting productive capacity to US-bound exports in affected product categories — depends critically on their structural productive capabilities. High-ECI countries are hypothesized to do so more successfully. Second, countries embedded in supply chains that include both US and Chinese nodes as buyers and suppliers face disruption to established trade flows, requiring reconfiguration. High-ECI countries are hypothesized to reconfigure more effectively.

3.2 | Sample and Data Sources

Our analysis covers a balanced panel of 130 countries over the period 2012–2022, yielding 1,430 country-year observations. The sample includes all countries for which complete data are available across the key data sources, subject to minimum data quality requirements described in Appendix A. The pre-shock period (2012–2017) provides six years of baseline observations; the post-shock period (2018–2022) covers the tariff implementation and its aftermath, including the COVID-19 disruptions of 2020, which we address in robustness checks through the inclusion of year fixed effects and a COVID indicator.

Data on ECI, PCI, and COI are drawn from the Harvard Growth Lab's Atlas of Economic Complexity (version 10.0; 2024), which provides annual estimates for these indices from 1995 onward based on reconciled bilateral trade data from UN Comtrade. Forward and backward GVC linkage shares are derived from the OECD Trade in Value Added (TiVA) 2023 dataset, which provides annual estimates of GVC participation at the country level. Bilateral trade flow data for the construction of the Export Recovery Index and the Partner Diversification Index are obtained from UN Comtrade via the BACI database (Gaulier & Zignago, 2010), which harmonizes bilateral trade data to resolve reporting discrepancies. Bilateral tariff data at the HS-6 level are sourced from the UNCTAD TRAINS database as compiled in the World Bank's WITS platform. Country-level controls are drawn from the World Bank's World Development Indicators and the World Bank Worldwide Governance

Indicators. The Geopolitical Risk Index is obtained from Caldara and Iacoviello (2022), which provides monthly GPR estimates back to 1985, aggregated to annual averages for this study.

3.3 | Variable Operationalization

3.3.1 | Dependent Variables

Dependent Variable 1 — GVC Linkage Stability (Δ Fwd. Linkage): Measured as the annual change in a country's forward GVC linkage share, defined following Johnson and Noguera (2012) as the share of domestic value added in a country's gross exports that is ultimately absorbed in a third country. This measure captures the degree to which a country's productive capacity is integrated into downstream global value chains. A smaller (less negative) change following the tariff shock indicates greater GVC linkage stability.

Dependent Variable 2 — Export Recovery Index (ERI): A composite measure constructed as follows. For each country and each year in the post-shock period, we compute the ratio of total export value (in constant 2015 USD, from UN Comtrade/BACI) to the country's average pre-shock export value over 2015–2017. This ratio captures the degree to which a country has recovered its export volumes relative to the pre-shock baseline. Values above 1 indicate full recovery (or growth beyond pre-shock levels); values below 1 indicate incomplete recovery. The ERI ranges from 0 to approximately 2 in our sample. Full construction details are in Appendix C.

Dependent Variable 3 — Trade Partner Diversification ($1 - \text{HHI}$): The inverse of the Herfindahl-Hirschman Index (HHI) computed over a country's export destination shares in a given year. Higher values indicate greater diversification across trading partners. We use the inverse HHI so that a higher value of the dependent variable consistently indicates a more resilient outcome. This measure captures the degree to which a country successfully maintains or broadens its trade partner base under shock conditions.

3.3.2 | Independent and Moderating Variables

Economic Complexity Index (ECI): The primary independent variable, drawn directly from the Harvard Growth Lab Atlas of Economic Complexity. The ECI is standardized to mean zero and unit standard deviation across the country-year sample. We use the most recently released ECI estimates for each year, incorporating the Growth Lab's updated trade data reconciliation methodology (Bustos et al., 2026).

Complexity Outlook Index (COI): The moderating variable capturing product space density, also from the Harvard Growth Lab Atlas, standardized analogously to ECI.

Geopolitical Risk Index (GPR): The country-year measure of geopolitical risk from Caldara and Iacoviello (2022), standardized to mean zero and unit standard deviation in regressions.

Tariff Shock Exposure ($\text{Post} \times \text{Exposed}$): A binary treatment variable interacted with the post-period indicator. Exposed countries are defined as those in the top tertile of bilateral US–China trade intensity — measured as the sum of US and Chinese shares in a country's total trade — averaged over the pre-shock period 2015–2017. Post is a binary indicator equal to one for years 2019–2022. The triple interaction $\text{ECI} \times \text{Post} \times \text{Exposed}$ is the key identifying variation in our main specifications.

3.3.3 | Control Variables

We include a set of time-varying country-level controls motivated by prior literature on trade resilience and GVC participation: (1) log GDP per capita, controlling for overall development level (Mankiw et al., 1992); (2) trade openness as a share of GDP, controlling for baseline trade integration; (3) an institutional quality composite index from the World Governance Indicators (average of six WGI dimensions), controlling for governance quality that may independently shape trade capacity; (4) natural resource rents as a share of GDP, controlling for resource-dependence which is both correlated with ECI and negatively associated with trade diversification; and (5) bilateral US–China trade intensity (described above), included as a main effect to control for systematic differences between exposed and non-exposed countries. All continuous controls are standardized in the regression models.

4 | RESULTS

4.1 | Descriptive Statistics

Table 1 presents descriptive statistics for all variables. The ECI ranges from -2.89 (lowest complexity, e.g., commodity-exporting developing economies) to 2.78 (highest complexity; Japan, South Korea, Switzerland), with a mean near zero consistent with standardization. Substantial heterogeneity is evident across all three dependent variables: the standard deviation of the GVC linkage stability measure (0.183) relative to its mean (-0.024) indicates considerable cross-country variation in the magnitude of disruption experienced. The ERI mean of 0.683 indicates that, on average, countries recovered only 68% of their pre-shock export volumes over the post-period — underscoring the severity of the disruption — but with high variance ($\text{SD} = 0.312$) suggesting that recovery trajectories diverge substantially across countries. Correlation analysis (Appendix D, Table D1) reveals that ECI is positively correlated with all three resilience dimensions ($r = 0.39, 0.44, 0.37$ respectively), providing preliminary support for the hypotheses, while VIF values uniformly below 2.6 indicate that multicollinearity does not threaten inference.

TABLE 1 Descriptive Statistics and Variable Summary ($N = 130$ Countries, 2012–2022)

Variable	N	Mean	SD	Min	Max	VIF
<i>Dependent Variables</i>						
GVC linkage stability (Δ Fwd. linkage)	1,430	-0.024	0.183	-0.612	0.489	—
Export recovery index (ERI)	1,430	0.683	0.312	0.000	1.000	—
Partner diversification ($1 - \text{HHI}$)	1,430	0.421	0.278	0.014	0.951	—
<i>Independent Variables</i>						
Economic Complexity Index (ECI)	1,430	0.118	0.984	-2.893	2.776	1.82
Complexity Outlook Index (COI)	1,430	-0.178	1.124	-3.211	2.945	1.94
Product space density (relatedness)	1,430	0.342	0.143	0.071	0.821	2.11
Tariff shock exposure (Post $\times$ Exposed)	1,430	0.231	0.422	0.000	1.000	1.63
<i>Moderating Variables</i>						
Geopolitical risk (GPR index)	1,430	92.43	47.28	18.72	411.63	1.47
<i>Control Variables</i>						
GDP per capita (log)	1,430	9.138	1.241	5.894	11.643	2.38
Trade openness (% GDP)	1,430	87.62	52.34	21.44	441.6	2.17
Institutional quality (WGI composite)	1,430	0.024	0.981	-2.134	1.983	2.54
Natural resource rents (% GDP)	1,430	8.743	12.18	0.000	68.43	1.89
US-China bilateral trade intensity	1,430	0.183	0.211	0.001	0.894	2.03

Note: $N = 130$ countries $\times$ 11 years (2012–2022). VIF = variance inflation factor. All continuous variables are standardized in regression models. HHI = Herfindahl–Hirschman Index (lower = more diversified; reversed so higher = more diversified). GVC = global value chain. GPR index from Caldara and Iacoviello (2022). ECI, COI, and product space data from Harvard Growth Lab Atlas of Economic Complexity (2024).

4.2 | Main Effects: H1 – GVC Linkage Stability

Table 2 presents results for Hypothesis 1. Model 1 estimates the baseline triple-difference specification without controls; Model 2 adds the full set of time-varying controls; Model 3 is our preferred full specification; and Model 4 estimates the propensity score matched DiD. Across all four specifications, the coefficient on the key triple interaction $\text{ECI} \times \text{Post} \times \text{Exposed}$ is positive, statistically significant, and substantively meaningful. In our preferred specification (Model 3), $\beta_3 = 0.092$ ($\text{SE} = 0.021, p < 0.01$), indicating that a one-standard-deviation increase in ECI is associated with a 0.092-unit smaller annual decline in forward GVC linkage share among countries exposed to the tariff shock during the post-period, relative to unexposed countries. Given that the mean disruption to GVC linkages in exposed countries is approximately -0.14 percentage points, this coefficient implies that moving from the 25th to the 75th percentile of ECI (approximately a 1.3 SD increase) reduces disruption by approximately 83% of its mean magnitude – a substantial buffering effect. Results are virtually unchanged in the PSM-DiD specification (Model 4; $\beta_3 = 0.084$), and the pre-period $\text{Post} \times \text{Exposed}$ main effect is negative and significant ($\beta = -0.091$), consistent with tariff shocks generating real disruptions that ECI partially offsets. H1 is supported.

4.3 | Main Effects: H2 – Export Portfolio Recovery Speed

Table 3 presents results for Hypothesis 2. The pattern of results is consistent with and stronger than those for H1. In the full specification (Model 3), $\beta_3 = 0.161$ ($\text{SE} = 0.034, p < 0.01$). A one-standard-deviation increase in ECI is associated with a 0.161-unit improvement in the Export

Recovery Index among tariff-exposed countries during the post-period. Given that the mean ERI for the post-period in exposed countries is approximately 0.54 (compared with 0.74 for non-exposed countries), this coefficient implies that the highest-ECI exposed countries (top quartile, $\text{ECI} \approx +1.5$ SD) recover to near-full pre-shock levels, while the lowest-ECI exposed countries (bottom quartile, $\text{ECI} \approx -1.4$ SD) recover less than 40% of pre-shock volumes. The magnitude of this effect underscores the economically important role of structural national complexity in conditioning export recovery. H2 is supported.

4.4 | Main Effects: H3 – Trade Partner Diversification

Table 4 presents results for Hypothesis 3. In the full specification (Model 3), $\beta_3 = 0.114$ ($\text{SE} = 0.037, p < 0.01$). High-ECI countries facing tariff shock exposure successfully diversify their trade partners to a significantly greater degree than low-ECI exposed countries, consistent with H3. The effect size is somewhat smaller than for DV2, consistent with the theoretical expectation that trade partner diversification is a more challenging and longer-horizon resilience response than export volume recovery – it requires not merely restoring existing relationships but building new ones. Nonetheless, the effect is substantively important: moving from the median to the 75th percentile of ECI is associated with a 0.057-unit increase in the ($1 - \text{HHI}$) partner diversification index, representing approximately a 14% improvement relative to the mean. H3 is supported.

4.5 | Moderation Analysis: H4 and H5

Table 5 presents the moderation results. For H4, we find that the Complexity Outlook Index (COI) positively and significantly moderates the

TABLE 2 Two-Way Fixed Effects DiD Estimates: Effect of ECI on GVC Linkage Stability (DV1)

	Model 1 Baseline	Model 2 + Controls	Model 3 Full DiD	Model 4 PSM-DiD
ECI	0.038* (0.021)	0.041* (0.022)	0.044** (0.018)	0.039* (0.020)
Post × Exposed (PE)	-0.098*** (0.019)	-0.094*** (0.018)	-0.091*** (0.016)	-0.089*** (0.017)
ECI × PE (H1)	0.089*** (0.024)	0.087*** (0.023)	0.092*** (0.021)	0.084*** (0.022)
GDP per capita (log)		0.021** (0.010)	0.019** (0.009)	0.018* (0.010)
Trade openness		0.014* (0.008)	0.012 (0.008)	0.011 (0.009)
Institutional quality (WGI)		0.031*** (0.009)	0.029*** (0.008)	0.027*** (0.009)
Natural resource rents		-0.022** (0.010)	-0.019** (0.009)	-0.021** (0.010)
US-China trade intensity			-0.044*** (0.013)	-0.041*** (0.014)
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Constant	-0.041*** (0.012)	-0.089*** (0.021)	-0.083*** (0.019)	-0.078*** (0.020)
Observations	1,430	1,430	1,430	1,218
R ² (within)	0.214	0.261	0.289	0.281

Note: Dependent variable = annual change in OECD TiVA forward GVC linkage share. Two-way fixed effects panel estimator. Standard errors clustered by country in parentheses. PSM-DiD = propensity score matched difference-in-differences. PE = Post × Exposed interaction (Post = 2019–2022; Exposed = countries in top tertile of bilateral US–China trade intensity). *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

TABLE 3 Two-Way Fixed Effects DiD Estimates: Effect of ECI on Export Recovery Index (DV2)

	Model 1 Baseline	Model 2 + Controls	Model 3 Full DiD	Model 4 PSM-DiD
ECI	0.064** (0.029)	0.069** (0.028)	0.072*** (0.025)	0.068** (0.027)
Post × Exposed (PE)	-0.142*** (0.028)	-0.137*** (0.027)	-0.133*** (0.024)	-0.129*** (0.026)
ECI × PE (H2)	0.156*** (0.038)	0.153*** (0.036)	0.161*** (0.034)	0.148*** (0.036)
GDP per capita (log)		0.037*** (0.013)	0.034*** (0.012)	0.031*** (0.013)
Trade openness		0.022** (0.011)	0.019* (0.010)	0.017* (0.010)
Institutional quality (WGI)		0.044*** (0.013)	0.040*** (0.012)	0.038*** (0.013)
Natural resource rents		-0.033*** (0.012)	-0.029** (0.011)	-0.027** (0.013)
US-China trade intensity			-0.061*** (0.018)	-0.058*** (0.019)
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Constant	0.612*** (0.018)	0.481*** (0.031)	0.497*** (0.029)	0.512*** (0.031)
Observations	1,430	1,430	1,430	1,218
R ² (within)	0.241	0.298	0.324	0.317

Note: Dependent variable = Export Recovery Index (ERI); see Appendix C for construction. Two-way fixed effects panel estimator with country-clustered standard errors. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

ECI-resilience relationship across all three dependent variables. In the GVC stability specification, the triple-difference interaction $ECI \times PE \times COI = 0.067$ (SE = 0.022, $p < 0.01$); for export recovery, $ECI \times PE \times$

$COI = 0.091$ (SE = 0.033, $p < 0.01$); and for partner diversification, $ECI \times PE \times COI = 0.074$ (SE = 0.031, $p < 0.05$). These results indicate that the product-space density captured by COI amplifies the resilience

TABLE 4 Two-Way Fixed Effects DiD Estimates: Effect of ECI on Trade Partner Diversification (DV3)

	Model 1 Baseline	Model 2 + Controls	Model 3 Full DiD	Model 4 PSM-DiD
ECI	0.049* (0.026)	0.052** (0.025)	0.055** (0.023)	0.051* (0.026)
Post × Exposed (PE)	-0.119*** (0.024)	-0.113*** (0.023)	-0.109*** (0.021)	-0.104*** (0.022)
ECI × PE (H3)	0.112*** (0.041)	0.108** (0.039)	0.114*** (0.037)	0.106** (0.040)
GDP per capita (log)		0.029** (0.012)	0.026** (0.011)	0.024* (0.013)
Trade openness		0.031*** (0.011)	0.028** (0.010)	0.026** (0.011)
Institutional quality (WGI)		0.038*** (0.012)	0.034*** (0.011)	0.033*** (0.012)
Natural resource rents		-0.041*** (0.013)	-0.037*** (0.012)	-0.035*** (0.013)
US-China trade intensity			-0.052*** (0.016)	-0.049*** (0.017)
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Constant	0.389*** (0.016)	0.284*** (0.028)	0.301*** (0.026)	0.318*** (0.028)
Observations	1,430	1,430	1,430	1,218
R ² (within)	0.198	0.247	0.271	0.263

Note: Dependent variable = (1 - HHI) trade partner diversification index (higher = more diversified). Two-way fixed effects panel estimator with country-clustered standard errors. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

advantage of productive complexity: countries that not only possess sophisticated current capabilities but also have many adjacent product pathways available for diversification achieve substantially better resilience outcomes. H4 is supported across all three dimensions.

For H5, the triple-difference interaction $ECI \times PE \times GPR = -0.043$ (SE = 0.018, $p < 0.05$) in the GVC stability specification, indicating that elevated geopolitical risk attenuates the ECI-resilience relationship. This is consistent with the theoretical argument that geopolitical barriers can close off the channels — global market access, buyer recognition, trade partner switching — through which productive complexity generates resilience. The attenuation effect is statistically significant but smaller in magnitude than the amplifying COI effect, suggesting that the structural resilience advantage of high ECI is robust to moderate geopolitical risk elevations but erodes under extreme conditions. H5 is supported.

4.6 | Robustness Checks

Table 6 presents a comprehensive battery of robustness checks. The placebo test — in which we falsely assign the tariff shock to 2014–2015 — yields coefficients near zero and statistically insignificant across all three DVs, strongly supporting the parallel trends assumption and ruling out pre-existing differential trends as an explanation for the main results. Alternative ECI threshold definitions (top quartile treatment vs. top tertile) produce virtually identical results. Excluding resource-rich countries (natural resource rents > 20% of GDP) and excluding OECD countries separately — addressing concerns that the ECI effect might be driven by high-income or resource-endowed country

sub-samples — leaves results essentially unchanged. Using lagged ECI ($t - 1$ rather than contemporaneous), stacked DiD event-study specifications, region-clustered standard errors, and Poisson pseudo-maximum-likelihood estimation (PPML, addressing potential issues with the log-linear approximation for trade data; Santos Silva & Tenreiro, 2006) all yield consistent results with coefficients ranging from 0.078 to 0.097 for DV1, 0.141 to 0.168 for DV2, and 0.099 to 0.119 for DV3. The stability of results across this range of alternative specifications substantially strengthens confidence in the main findings.

5 | DISCUSSION

5.1 | Theoretical Contributions

This paper advances supply chain management theory in three principal ways. First and most fundamentally, we challenge the firm-centric orthodoxy of supply chain resilience research by establishing a country-level structural construct — national productive complexity — as a significant, causally identified, and theoretically grounded determinant of supply chain resilience outcomes. Our results demonstrate that substantial heterogeneity in supply chain resilience across countries cannot be explained by firm-level management practices alone: structural national characteristics, specifically the depth and diversity of the national productive ecosystem as captured by the ECI, account for a sizeable and significant portion of cross-country variation in three distinct resilience dimensions. This finding extends the scope conditions of existing SCR

TABLE 5 Moderation Analysis: Product Space Density (H4) and Geopolitical Risk (H5)

	GVC Stability COI mod. (H4a)	Exp. Recovery COI mod. (H4b)	Ptr. Diversif. COI mod. (H4c)	GVC Stability GPR mod. (H5)
ECI × PE	0.088*** (0.022)	0.159*** (0.035)	0.111*** (0.038)	0.094*** (0.023)
Moderator (COI or GPR)	0.031*** (0.011)	0.047*** (0.017)	0.039** (0.016)	-0.018** (0.008)
ECI × PE × COI (H4a-c)	0.067*** (0.022)	0.091*** (0.033)	0.074** (0.031)	
ECI × PE × GPR (H5)				-0.043** (0.018)
All controls	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	1,430	1,430	1,430	1,430
R ² (within)	0.312	0.351	0.297	0.307

Note: Triple-difference specifications with full control set and two-way fixed effects. COI = Complexity Outlook Index (higher = more product-space opportunities adjacent to current capabilities). GPR = Geopolitical Risk Index (Caldara & Iacoviello, 2022; standardized). Standard errors clustered by country. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

TABLE 6 Robustness Check Summary: Key Triple-Interaction Coefficient (ECI × Post × Exposed) across Specifications

Specification	DV1 Coeff.	DV2 Coeff.	DV3 Coeff.	N
Main specification (Tables 2–4, M3)	0.092***	0.161***	0.114***	1,430
Placebo: Pre-period shock (2014–2015)	0.009 (0.019)	0.011 (0.027)	0.006 (0.031)	1,430
Alternative ECI threshold (top quartile)	0.097***	0.168***	0.119**	1,430
Exclude resource-rich countries (> 20% rents)	0.086***	0.155***	0.107**	1,118
Exclude OECD countries	0.081**	0.141***	0.098**	858
Lagged ECI ($t - 1$)	0.088***	0.157***	0.110**	1,300
Stacked DiD (event-study)	0.084***	0.152***	0.103**	1,430
Cluster SE by region	0.092***	0.161**	0.114**	1,430
Poisson pseudo-max. likelihood (PPML)	0.078***	0.144***	0.099**	1,430

Note: All specifications include full control set, country FE, and year FE. Coefficient shown is for the key triple interaction ECI × Post × Exposed (or analogue in PPML). Placebo test uses 2014–2015 as false tariff shock. Standard errors clustered by country unless noted. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

theory, which has been primarily developed and tested in firm-level or dyadic contexts. Second, we contribute to supply chain management theory by identifying the theoretical mechanism through which national structure shapes resilience. Our RBV-grounded argument — that national productive complexity constitutes a national-level resource endowment that is valuable, rare, inimitable, and non-substitutable, and that this endowment enables the firm-level resilience mechanisms emphasized in extant theory — provides a theoretically coherent bridge between macro-level economic structure and firm-level supply chain behavior. This bridging argument challenges future SCR researchers to consider more carefully the structural environments in which their firm-level theories operate, and suggests that the apparent explanatory power of firm-level resilience constructs may partially reflect national structural context that has been omitted from their analyses.

Third, we establish the theoretical relevance of economic complexity theory for supply chain management. The ECI, COI, and Product Space concepts have been developed and applied extensively in economic geography and development economics, but have not previously

been incorporated into SCM theory. Our results suggest that these constructs are powerful predictors of supply chain resilience outcomes, and that the theoretical infrastructure of complexity economics — productive knowhow, capability adjacency, product space navigation — offers valuable conceptual resources for advancing understanding of supply chain adaptation and reconfiguration under disruption. This opens a productive interdisciplinary dialogue between SCM and complexity economics.

Importantly, our theoretical contribution does not supersede or contradict the firm-centric SCR literature; it contextualizes and enriches it. Firms with superior internal capabilities, stronger relational assets, and more sophisticated supply chain management practices will achieve better resilience outcomes within any national context. But our results suggest that these firm-level advantages are more effective — and more easily achieved — when firms are embedded in nationally complex productive ecosystems. The national structural context is the substrate in which firm-level resilience mechanisms are grown and from which they draw part of their power. Future SCM theory needs to account for both levels of analysis.

5.2 | Managerial Implications

Our findings carry direct and actionable implications for supply chain managers grappling with the tariff and geopolitical risk environment of the mid-2020s. Most immediately, the ECI — freely available from the Harvard Growth Lab's Atlas of Economic Complexity — provides supply chain strategists with a validated, theoretically grounded macro-level screening criterion for evaluating the resilience profile of sourcing country candidates. Our results suggest that, all else equal, sourcing from higher-ECI countries provides structural insurance against the supply chain disruptions generated by tariff escalation and geopolitical friction: suppliers in complex economies are better positioned to recover, adapt, and diversify their own supply arrangements when shocks hit.

This implication is particularly relevant for firms currently executing China+1 or friend-shoring supply chain diversification strategies. The conventional criteria applied to alternative sourcing locations — labor costs, infrastructure quality, proximity, existing trade agreements — should be supplemented with ECI-based assessment of structural resilience capacity. A country with competitive labor costs and adequate infrastructure but low ECI (such as many commodity-dependent developing economies) may prove substantially less resilient as an alternative sourcing location under future shocks than a higher-cost but more complex economy. The ECI provides a forward-looking structural assessment that complements point-in-time cost analysis.

The moderation results provide additional practical guidance. The COI — also freely available — should be used to assess not only a country's current resilience profile but its adaptive trajectory: countries with high COI are positioned to diversify their productive capabilities rapidly in response to demand signals, suggesting that they may improve their resilience profiles quickly even if current ECI is moderate. Conversely, the GPR moderation result counsels supply chain managers to recognize that structural productive complexity provides limited resilience protection under extreme geopolitical confrontation: in severely geopolitically contested environments, complexity advantages may be substantially negated by market access restrictions and sanctions regimes.

5.3 | Policy Implications

For policymakers, our findings provide robust empirical support for productive diversification strategies as a supply chain resilience investment. Countries seeking to improve the resilience of their domestic supply chains to external trade shocks should prioritize policies that accelerate economic diversification and productive complexity growth — industrial policies that encourage entry into adjacent product spaces, investments in education and technical training that build the distributed productive knowhow captured by the ECI, and trade facilitation measures that support small and medium enterprises' access to global markets. Our results suggest that such investments deliver a supply chain resilience dividend that extends beyond their direct economic development benefits.

Importantly, our results also suggest a policy sequencing insight: the COI moderation result indicates that countries with many adjacent product-space opportunities (high COI) have a higher return to complexity-building investments, because the structural conditions for productive diversification are already partially in place. Policymakers in these countries can achieve faster ECI growth — and thus faster resilience improvements — through targeted support for capability expansion into COI-identified adjacent products than through more diffuse horizontal interventions.

5.4 | Limitations and Future Research

This study has several important limitations that suggest avenues for future research. First, while our DiD design with parallel trends validation and extensive robustness checks provides strong evidence for causal identification, the natural experiment design necessarily relies on assumptions — particularly the parallel trends assumption and the exogeneity of tariff shock exposure — that cannot be fully verified. Future research using alternative natural experiments or exogenous variation in ECI itself (such as large-scale shocks to specific capability sectors in particular countries) could provide additional causal validation.

Second, our analysis operates at the country level, which necessarily abstracts from firm-level and sector-level heterogeneity within countries. The ECI-resilience relationship we document at the country level may mask important within-country variation: firms in high-ECI sectors within low-ECI countries, or firms with globally sophisticated operations embedded in domestically simple economies, may exhibit resilience profiles that deviate from our country-level predictions. Future research should examine the micro-foundations of the country-level effects we document, potentially using firm-level panel data matched to country-level ECI measures.

Third, our focus on the US–China tariff war as the primary natural experiment, while offering strong identification advantages, limits generalizability to tariff-induced shocks specifically. The mechanisms we propose should be relevant to other types of supply chain disruption — pandemics, natural disasters, geopolitical conflicts — but the magnitude and directionality of the ECI effect may vary across disruption types. Future research should examine whether the structural resilience advantages of productive complexity extend to non-tariff disruptions, and whether the mechanisms we identify operate differently under supply-side versus demand-side shocks.

Fourth, the time horizon of our study (2012–2022) means that some resilience processes — particularly long-run structural transformation of national productive capabilities in response to sustained trade disruption — may not be fully captured. Longitudinal extensions examining the 2025 tariff escalation and its aftermath, as data become available, will provide further tests of our theoretical framework in an even more severe disruption context.

6 | CONCLUSIONS

This paper establishes national productive complexity as a structural macro-level determinant of supply chain resilience under tariff and geopolitical shocks, challenging the field's dominant firm-centric paradigm and opening a new theoretical frontier for supply chain management research. Using a natural experiment based on the 2018–2019 US–China trade war and a panel dataset of 130 countries over 2012–2022, we demonstrate that the ECI — a composite measure of the diversity and sophistication of a country's productive capabilities — causally predicts three dimensions of supply chain resilience: GVC linkage stability, export portfolio recovery, and trade partner diversification under stress. These effects are amplified by product-space density and attenuated under extreme geopolitical risk, consistent with the theoretical mechanisms we propose. The implications of this finding extend across theory, management practice, and public policy. For SCM theory, it mandates a broadening of the unit of analysis to encompass the national structural contexts in which firm-level resilience mechanisms operate. For supply chain practitioners, it offers a theoretically validated, publicly available macro-level indicator — the ECI — for assessing the structural resilience profile of sourcing countries under the new tariff and geopolitical risk landscape. And for policymakers, it provides causal evidence that investments in productive diversification and economic complexity growth generate supply chain resilience dividends at the national level. As the global economy enters a period of sustained tariff volatility and geopolitical realignment, the question of which countries and supply chain nodes can absorb, adapt to, and recover from structural trade disruptions will become increasingly decisive for global production networks. This paper provides a theoretical framework and empirical evidence for understanding why some nations — and the supply chains embedded within them — are structurally better equipped to survive what lies ahead.

AUTHOR CONTRIBUTIONS

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FINANCIAL DISCLOSURE

None reported.

CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

REFERENCES

SUPPORTING INFORMATION

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APPENDIX

A COUNTRY SAMPLE

The analysis includes 130 countries for which complete data are available across all key data sources over the period 2012–2022. Countries are excluded if they have (1) fewer than five years of non-missing data across the core variables, (2) populations below one million (small island states and micro-states, for which trade data are unreliable), or (3) extreme commodity concentration with natural resource rents exceeding 90% of GDP in any year of the sample period (nine countries excluded on this criterion: Brunei, Equatorial Guinea, Kuwait, Libya, Qatar, Saudi Arabia, South Sudan, Timor-Leste, and Turkmenistan). The resulting sample includes 53 high-income countries, 42 upper-middle-income countries, and 35 lower-middle or low-income countries (World Bank income classification, 2022), providing substantial variation in ECI and other key variables.

ECI rankings in the sample range from Japan (rank 1, ECI = 2.78) and South Korea (rank 2, ECI = 2.44) at the high end, to Mauritania (rank 127, ECI = −2.73) and Guinea (rank 128, ECI = −2.81) at the low end. The top quartile of ECI countries (ECI > 1.0) includes primarily OECD economies plus a number of East and Southeast Asian manufacturing hubs (e.g., Taiwan, Singapore, Thailand). The bottom quartile (ECI < −1.0) consists primarily of commodity-exporting developing economies in Sub-Saharan Africa, Central Asia, and the Middle East. The middle two quartiles contain considerable heterogeneity, including rapidly diversifying emerging economies (Vietnam, Bangladesh, Mexico, Turkey) that are highly relevant to current supply chain reconfiguration strategies.

B US–CHINA TARIFF SHOCK TIMELINE

The US–China tariff escalation unfolded across four staggered lists, creating multiple waves of trade disruption that we use for identification:

List 1 (July 6, 2018): \$34 billion in Chinese goods, primarily industrial machinery, electronic components, and aerospace equipment. Initial tariff rate: 25%. This round targeted products identified in the USTR Section 301 investigation as benefiting from Chinese forced technology transfer and IP theft practices.

List 2 (August 23, 2018): \$16 billion in additional Chinese goods, including semiconductors, chemicals, and plastics. Initial tariff rate: 25%. This round expanded the scope of the Section 301 action.

List 3 (September 24, 2018; escalated May 10, 2019): \$200 billion in Chinese goods, including a wide range of consumer products, furniture, machinery, and food products. Initial tariff rate: 10%, escalated to 25% in May 2019. This was the quantitatively largest single tariff action and had the broadest supply chain implications.

List 4 (September 1, 2019): \$112 billion in Chinese goods, including consumer electronics and apparel. Initial tariff rate: 7.5%. (List 4B was

suspended and subsequently canceled under the Phase One Agreement of January 2020.)

The staggered timing of these lists, covering different product categories at different rates and at different points in time, provides additional variation within our DiD design that we exploit in the stacked DiD event-study specifications reported in Table 6.

C VARIABLE CONSTRUCTION DETAILS

For each country i and post-shock year t (2019–2022),

$$ERI_{it} = \frac{\text{ExportValue}_{it}}{\text{mean}(\text{ExportValue}_{i, 2015-2017})},$$

where ExportValue is total goods export value in constant 2015 USD, sourced from the BACI database. Values are winsorized at the 1st and 99th percentiles to reduce the influence of extreme outliers (primarily driven by entrepôt trade effects in small trading hubs). Countries with export value growth above 200% of pre-shock baseline are flagged for sensitivity analysis.

Trade Partner Diversification (1–HHI) Construction: For each country i and year t ,

$$HHI_{it} = \sum_j s_{ijt}^2,$$

where s_{ijt} is the share of country i 's total exports destined for partner country j in year t , computed from BACI bilateral trade data. We include all destination countries with non-zero bilateral trade flows. The measure is inverted (1–HHI) so that higher values indicate greater diversification. The index ranges from near zero (extreme concentration on a single partner) to approximately 1 (maximum feasible diversification).

US–China Bilateral Trade Intensity: For each country i and pre-shock period, we compute the sum of the US share and Chinese share in country i 's total bilateral trade (exports + imports), averaged over 2015–2017. Countries in the top tertile of this measure are classified as 'Exposed' for the DiD treatment definition. Sensitivity to alternative treatment definitions (top quartile, continuous exposure intensity) is reported in Table 6.

D SUPPORTING ANALYSES

Table D1 – Correlation Matrix: Full pairwise Pearson correlations among all study variables. ECI is positively correlated with all three DVs ($r = 0.39, 0.44, 0.37$ for DV1, DV2, DV3 respectively) and with COI ($r = 0.61$). Maximum pairwise correlation in the control set is 0.58 (GDP per capita, institutional quality), well below the 0.80 threshold conventionally used to identify problematic multicollinearity. All correlations are consistent with the theoretical predictions and do not suggest multicollinearity concerns in the regression models. Parallel Trends Test: We estimate an event-study specification including leads and lags of the Post $\times$ Exposed treatment indicator, interacted with ECI, to examine whether the ECI-resilience relationship diverges between exposed and non-exposed countries prior to the 2018 tariff shock. Results (Figure D1, available upon request) show statistically insignificant pre-trend coefficients for all three dependent variables in the five years preceding the shock (2013–2017), supporting the parallel trends assumption. Post-shock coefficients (2018–2022) are uniformly positive and increasing

in magnitude for ECI $\times$ Post $\times$ Exposed, consistent with a sustained buffering effect of ECI.

COVID-19 Sensitivity: The 2020 COVID-19 pandemic represents a major confounding event in the post-shock period. We address this in three ways: (1) year fixed effects absorb global year-specific shocks, including the average global COVID effect in 2020; (2) we estimate alternative specifications excluding 2020 from the post-period, finding results substantively unchanged ($\beta_3 = 0.087, 0.154, 0.109$ for DV1, DV2, DV3); and (3) we include a country-level COVID severity indicator (excess mortality per million, sourced from Our World in Data) as an additional control, finding results again largely unchanged. These analyses confirm that the ECI-resilience relationship is not driven by COVID-19 confounding.

AUTHOR BIOGRAPHY

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